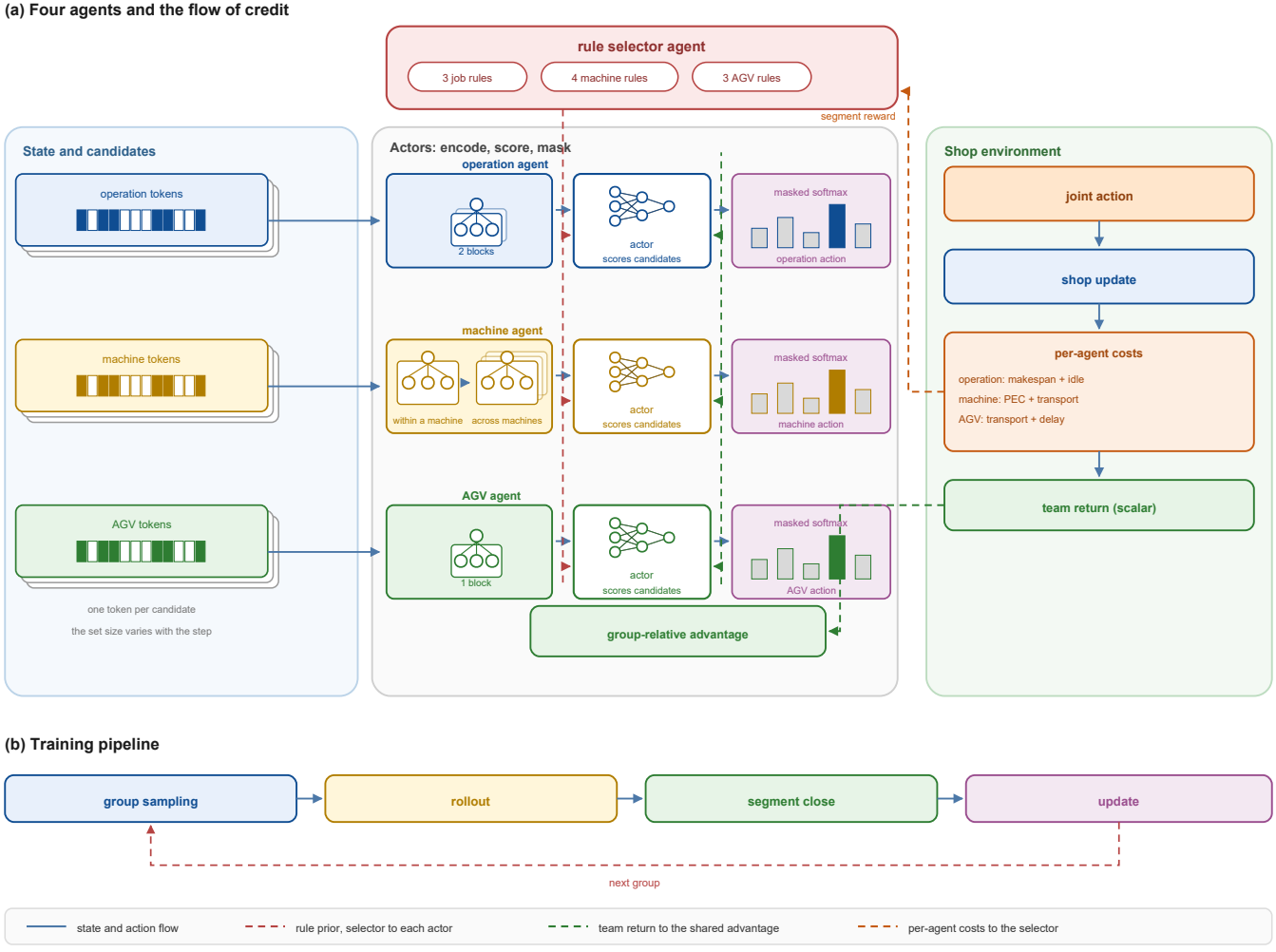
(a) Four agents and the flow of credit
rule selector agent
3 job rules
4 machine rules
3 AGV rules
segment reward
State and candidates
operation tokens
machine tokens
AGV tokens
one token per candidate
the set size varies with the step
Actors: encode, score, mask
operation agent
2 blocks
actor
scores candidates
masked softmax
operation action
machine agent
within a machine
across machines
actor
scores candidates
masked softmax
machine action
AGV agent
1 block
actor
scores candidates
masked softmax
AGV action
group-relative advantage
Shop environment
joint action
shop update
per-agent costs
operation: makespan + idle
machine: PEC + transport
AGV: transport + delay
team return (scalar)
(b) Training pipeline
group sampling
rollout
segment close
update
next group
state and action flow
rule prior, selector to each actor
team return to the shared advantage
per-agent costs to the selector